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Architecture of Spinnaker API

Spinnaker API is built around the GenICam standard, which offers a generic programming interface for various cameras and interfaces. Spinnaker is an extension of GenAPI. Spinnaker provides quick and easy access to your camera.

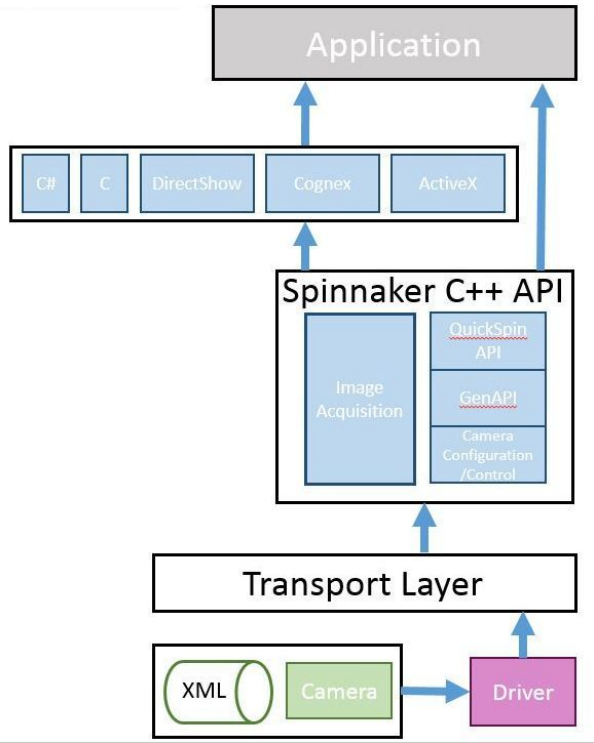
Spinnaker API includes two major components:

Image Acquisition

This is the acquisition engine that is responsible for setting up image buffers and image grabbing.

Camera Configuration

This is the configuration engine that is responsible for controlling your camera. This component consist of quick spin API, which is a wrapper that makes GenAPI easy to use.



Examples

Included with the Spinnaker SDK are a number of source code examples to help you get started. These examples are provided for C, C++, C#, and VB.NET languages and are precompiled for your convenience.

The table below describes the available Spinnaker SDK examples.

Spinnaker Example	Description
Acquisition	Enumerate, start acquisition, and grab images
AcquisitionMultipleCamera	How to capture images from multiple cameras simultaneously
ChunkData	How to get chunk data on an image, either from the nodemap or from the image itself

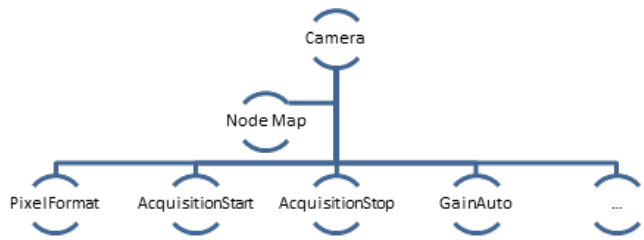
*Also available in QuickSpin

Spinnaker Example	Description
DeviceEvents	Create a handler to access device events
Enumeration*	Enumerate interfaces and cameras
EnumerationEvents	Explore arrival and removal events on interfaces and the system
Exposure*	Configure a custom exposure time
ImageEvents	Image events shows how to acquire images using the image event handler.
ImageFormatControl*	Configure a custom image size and format
Logging	Create a logging event handler
LookupTable	Configure lookup tables for the customization and control of individual pixels
NodeMapCallback	Create, register, use, and unregister callbacks
NodeMapInfo	How to retrieve node map information
SaveToAVI	Save images in AVI format
Sequencer (Blackfly S only)	Capture multiple images with different parameters in a sequence
SpinSimpleGUI_MFC	Graphical User Interface for evaluating and setting camera parameters
Trigger*	Trigger shows how to trigger the camera.

*Also available in QuickSpin

Nodes

Every GenICam compliant camera has an XML description file. The XML describes camera features, their interdependencies, and all other information like availability, access control, and minimum and maximum values. These features include Gain, Exposure Time, Image Format, and others. The elements of a camera description file are represented as software objects called Nodes. A Node map is a list of nodes created dynamically at run time.



To access camera properties such as setting image width:

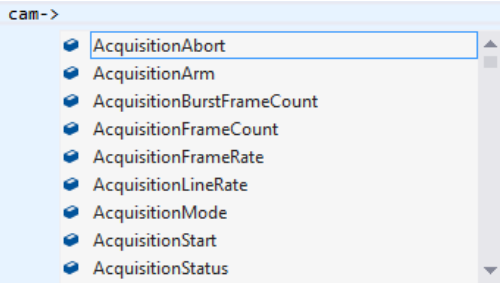
C++ GenAPI

```
GenApi::INodeMap & nodeMap = cam.GetNodeMap();
CIntegerPtr width = nodeMap.GetNode("Width");
width->SetValue(new_width_val);
```

QuickSpin API and Accessing Camera Parameters

Generic programming with GenICam requires developers to know feature names before using them. Spinnaker provides the QuickSpin API, which requires fewer lines of code and allows you to make use of auto completion. The QuickSpin API consists of a list of static functions integrated into the Camera class.

All camera parameters can be accessed through the camera pointer object.



Most camera parameters (all items in camera.h) can be accessed using the QuickSpin API.

For parameters not handled by QuickSpin API, you can access them via GenICam API (GenAPI). GenAPI is the generic programming interface for configuring all kinds of cameras. GenAPI is maintained by the European Machine Vision Association.

Below is an example comparison of inquiring camera gain via GenAPI and QuickSpin API.

C++ GenAPI

```
Spinnaker::GenApi::INodeMap & nodeMap = cam->GetNodeMap();
CFloatPtr GainNode = nodeMap.GetNode("Gain");
Float GainVal = GainNode->GetValue();
```

C++ QuickSpin API

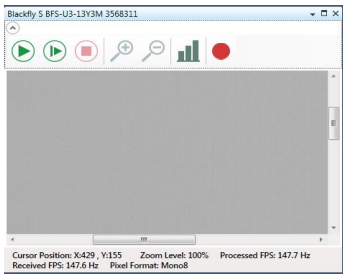
```
float quickGainVal = cam->Gain.GetValue();
```

C# Graphical User Interface API

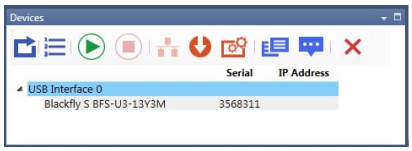
For applications that want to take advantage of Spinnaker's graphical user elements, graphical user interface (GUI) controls are available. GUI controls are divided into static and dynamic categories. Static GUI controls include the CameraSelectionDialog, display window, and property grid window. The GUI dynamically loads the camera's features from the firmware. Therefore, new firmware has the ability to add GUI controls to the same application, without recompiling.

Static GUI Dialogs

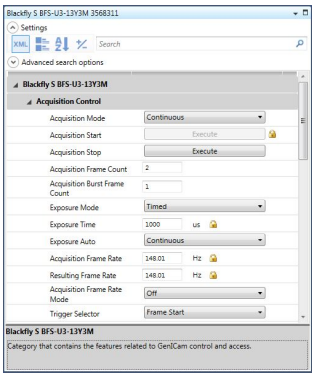
```
//To show image drawing window
GUIFactory AcquisitionGUI = new GUIFactory ();
AcquisitionGUI.ConnectGUILibrary(cam);
ImageDrawingWindow AcquisitionDrawing = AcquisitionGUI.GetImageDrawingWindow();
AcquisitionDrawing.Connect(cam);
AcquisitionDrawing.Start();
AcquisitionDrawing.ShowModal();
```




```
//To show camera selection window
GUIFactory AcquisitionGUI = newGUIFactory ();
AcquisitionGUI.ConnectGUILibrary(cam);
CameraSelectionWindow camSelection = AcquisitionGUI.GetCameraSelectionWindow();
camSelection.ShowModal(true);
```




```
//To show property grid window
GUIFactory AcquisitionGUI = new GUIFactory ();
AcquisitionGUI.ConnectGUILibrary(cam);
PropertyGridWindow propWindow = AcquisitionGUI.GetPropertyGridWindow();
propWindow.Connect(cam);
propWindow.ShowModal();
```



Dynamic GUI Control

```
GUIFactory dynamicGUI = new GUIFactory ();
dynamicGUI.ConnectGUILibrary(cam);
// Get dialog name via dynamicGUI.GetDialogNameList()
Window dlg = dynamicGUI.GetDialogByName(dialogName);
dlg.Owner = Window .GetWindow(this );
dlg.Show();
```

Camera XML

The camera's XML file contains information such as feature naming, register mapping, and dependencies between features. It is typical for GenICam-compliant software to cache the XML file for quicker access to the camera's definition. Spinnaker caches the XML file in a binary format to achieve better performance.

Camera XML files are located in:

C:\ProgramData\Spinnaker\XML

Recommended Environment

Spinnaker supports the following list of operating systems and development environments.

OS Compatibility (32- and 64-bit)	Windows XP Windows 7 Windows 8.1 Windows 10
Language Support	C C++ C# VB.NET
Compiler Support	Visual Studio 2010 Visual Studio 2013 Visual Studio 2015
Interface Support	USB3 Vision 1.0

Instantiate Cameras

Before you can instantiate a camera, you must create and initialize a system object. The System Singleton object is used to retrieve the list of interfaces (USB 3.1 or GigE) and cameras available. You must call ReleaseInstance() at the end of your program to free up the system object.

Multiple cameras can only be instantiated one at a time.

Instantiate multiple cameras (C++)

```
// Retrieve singleton reference to system object
SystemPtr system = System::GetInstance();

CameraList camList = system->GetCameras();
unsigned int numCameras = camList.GetSize();

for (unsigned int i = 0; i < numCameras; ++i)
{
  CameraPtr pCamera = cameraList.GetByIndex(i);

  pCamera->Init();

}

// Release system
system->ReleaseInstance();
```

Enumeration

The snippet below detects the number of cameras connected and enumerates them from an index.

Spinnaker C++ GenAPI

```
SystemPtr system = System::GetInstance();
CameraList camList = system->GetCameras();
unsigned int numCameras = camList.GetSize();
CameraPtr pCam = NULL;
for (int i = 0; i < numCameras; i++)
{
   pCam = camList.GetByIndex(i);
    pCam->Init();
}
```

Asynchronous Hardware Triggering

The snippet below does the following:

- Enables Trigger Mode
- Configures GPIO0/Line0 as the trigger input source
- Specifies the trigger signal polarity as an active high (rising edge) signal

Spinnaker C++ QuickSpin API

```
Cam->TriggerMode.SetValue(Spinnaker::TriggerModeEnums::TriggerMode_On);

Cam->TriggerSource.SetValue(Spinnaker::TriggerSourceEnums::TriggerSource_Line0);

Cam->TriggerSelector.SetValue(Spinnaker::TriggerSelectorEnums::TriggerSelector_FrameStart);

Cam->TriggerActivation.SetValue(Spinnaker::TriggerActivationEnums::TriggerActivation_RisingEdge);
```

Spinnaker C++ GenAPI

```
CEnumerationPtr triggerMode = nodeMap.GetNode("TriggerMode");
triggerMode->SetIntValue(triggerMode->GetEntryByName("On")->GetValue());

CEnumerationPtr triggerSource = nodeMap.GetNode("TriggerSource");
triggerSource->SetIntValue(triggerSource->GetEntryByName("Line0")->GetValue());

CEnumerationPtr triggerSelector = nodeMap.GetNode("TriggerSelector");
triggerSelector->SetIntValue(triggerSelector->GetEntryByName("FrameStart")->GetValue());

CEnumerationPtr triggerActivation = nodeMap.GetNode("TriggerActivation");
triggerActivation->SetIntValue(triggerActivation->GetEntryByName("RisingEdge")->GetValue());
```

Setting Black Level

BlackLevel is the GenICam feature that represents the DC offset that is applied to the video signal. This example compares the mechanism used to set this feature in both environments.

Spinnaker C++ QuickSpin API

```
// Brightness is called black level in GenICam
pCam->BlackLevelSelector.SetValue(Spinnaker::BlackLevelSelectorEnums::BlackLevelSelector_All);

//Set the absolute value of brightness to 1.5%.
pCam->BlackLevel.SetValue(1.5);
```

Spinnaker C++ GenAPI

```
CEnumerationPtr blackLevelSelector = nodeMap.GetNode("BlackLevelSelector");
blackLevelSelector->SetValue(SetIntValue(blackLevelSelector->GetEntryByName("True")->GetValue());

CFloatPtr blackLevel = nodeMap.GetNode("BlackLevel");
blackLevel->SetValue(1.5);
```

Setting Exposure Time

ExposureTime refers to the amount of time that the camera's electronic shutter stays open. This example sets your camera's exposure/shutter time to 20 milliseconds.

Spinnaker C++ QuickSpin API

```
// Turn off auto exposure
cam->ExposureAuto.SetValue(Spinnaker::ExposureAutoEnums::ExposureAuto_Off);

//Set exposure mode to "Timed"
cam->ExposureMode.SetValue(Spinnaker::ExposureModeEnums::ExposureMode_Timed);

//Set absolute value of shutter exposure time to 20000 microseconds
cam->ExposureTime.SetValue(20000);
```

Spinnaker C++ GenAPI

```
CEnumerationPtr exposureAuto = nodeMap.GetNode("ExposureAuto");
exposureAuto->SetIntValue(exposureAuto->GetEntryByName("Off")->GetValue());

CEnumerationPtr exposureMode = nodeMap.GetNode("ExposureMode");
exposureMode->SetIntValue(exposureMode->GetEntryByName("Timed")->GetValue());

CFloatPtr exposureTime = nodeMap.GetNode("ExposureTime");
exposureTime->SetValue(20000);
```

Setting Gain

The following code snippet adjusts gain to 10.5 dB.

Spinnaker C++ QuickSpin API

```
//Turn auto gain off
cam->GainAuto.SetValue(Spinnaker::GainAutoEnums::GainAuto_Off);

//Set gain to 10.5 dB
cam->Gain.SetValue(10.5);
```

Spinnaker C++ GenAPI

```
CEnumerationPtr gainAuto = nodeMap.GetNode("GainAuto");
gainAuto->SetIntValue(gainAuto->GetEntryByName("Off")->GetValue());

CFloatPtr gainValue = nodeMap.GetNode("Gain");
gainValue->SetValue(10.5);
```

Setting Gamma

The following code snippet adjusts gamma to 1.5.

Spinnaker C++ QuickSpin API

```
// Set the absolute value of gamma to 1.5
cam.Gamma.SetValue(1.5);
```

Spinnaker C++ GenAPI

```
CFloatPtr gamma = nodeMap.GetNode("Gamma");
gamma->SetValue(1.5);
```

Setting White Balance

The following code snippet adjusts the white balance's red and blue channels.

Spinnaker C++ QuickSpin API

```
//Set auto white balance to off
cam->BalanceWhiteAuto.SetValue(Spinnaker::BalanceWhiteAutoEnums::BalanceWhiteAuto_Off);

//Select blue channel balance ratio
cam->BalanceRatioSelector.SetValue(Spinnaker::BalanceRatioSelectorEnums::BalanceRatioSelector_Blue);

//Set the white balance blue channel to 2
CFloatPtr BalanceRatio = nodeMap.GetNode("BalanceRatio");
BalanceRatio->SetValue(2);
```

Spinnaker C++ GenAPI

```
//Set the white balance red channel to 2
cam->BalanceRatioSelector.SetValue(Spinnaker::BalanceRatioSelectorEnums::BalanceRatioSelector_Red);
BalanceRatio->SetValue(2);

CEnumerationPtr balanceWhiteAuto = nodeMap.GetNode("BalanceWhiteAuto");
balanceWhiteAuto->SetIntValue(balanceWhiteAuto->GetEntryByName("Off")->GetValue());

CEnumerationPtr balanceRatioSelector = nodeMap.GetNode("BalanceRatioSelector");
balanceRatioSelector->SetIntValue(balanceRatioSelector->GetEntryByName("Blue")->GetValue());

CFloatPtr balanceRatio = nodeMap.GetNode("BalanceRatio");
balanceRatio->SetValue(2);

balanceRatioSelector->SetIntValue(balanceRatioSelector->GetEntryByName("Red")->GetValue());
balanceRatio->SetValue(2);
```

Accessing Raw Bayer Data

Raw image data can be accessed programmatically via the getData method of the Spinnaker Image class. In 8 bits per pixel modes such as BayerRG8, the first byte represents the pixel at [row 0, column 0], the second byte at [row 0, column 1], and so on. The top left corner of the image data represents row 0, column 0.

Spinnaker C++ API

```
// Assuming image is 640 x 480 resolution. The current pixel format as well as PixelColorFilter indicate the Bayer Tile Mapping for the camera.
For example, BayerRG8 is RGGB.

ImagePtr pResultImage = cam.GetNextImage();
char* data = (char*)pResultImage->GetData();

// Assuming image is 640 x 480
// data[0] = Row 0, Column 0 = red pixel (R)
// data[1] = Row 0, Column 1 = green pixel (G)
// data[640] = Row 1, Column 0 = green pixel (G)
// data[641] = Row 1, Column 1 = blue pixel (B)
```

Setting Number of Image Buffers

The following code snippet adjusts the number of image buffers that the driver initializes for buffering images on your PC to 11 (default is 10).

Spinnaker C++ API

```
Spinnaker::GenApi::INodeMap & sNodeMap = cam->GetTLStreamNodeMap();
CIntegerPtr StreamNode = sNodeMap.GetNode("StreamDefaultBufferCount");
INT64 bufferCount = StreamNode->GetValue();
StreamNode->SetValue(11);
```

Event Handling

Spinnaker introduces two event classes: interface events and device events.

Interface Event

The interface event class is a new feature that is responsible for registering and deregistering user defined interface events such as device arrival and removal.

Interface Event C++

```
class InterfaceEventsHandler : public InterfaceEvent
{

    public :
    InterfaceEventsHandler(){};
    virtual ~InterfaceEventsHandler(){};

    void OnDeviceArrival()
    {

        std::cout<< "A Camera Arrived" << std::endl;

    };

    void OnDeviceRemoval( uint64_tdeviceSerialNumber )
    {
```

```
            std::cout<< "A Camera was removed with serial number: " <<
            deviceSerialNumber << std::endl;

        };

    };

    InterfaceEventsHandler handler;
    cam->RegisterEvent(handler);
```

Device Event

The device event class is responsible for registering and deregistering user defined device events such as start or end of exposure.

Device Event C++

```
// Select the Exposure End event
Spinnaker::GenApi:: CEnumerationPtr pEnum = nodeMap .GetNode( "EventSelector" );
pEnum->SetIntValue(pEnum->GetEntryByName( "EventExposureEnd" )->GetValue());

// Turn on the Event notification for Exposure End Event
Spinnaker::GenApi:: CEnumerationPtr pBool = nodeMap .GetNode( "EventNotification" );
pBool->SetIntValue(1);

// Once Exposure End Event is detected, the OnDeviceEvent function will be called
classDeviceEventHandler : publicDeviceEvent
{

    public :
    DeviceEventHandler(){};
    ~DeviceEventHandler(){};

    void OnDeviceEvent( Spinnaker::GenICam::gcstring eventNameeventId )
    {

        std::cout << "Got Device Event with " << eventName << " and ID=" << GetDeviceEventId() << std::endl;

    }

};

// Register event handler
DeviceEventHandler allDeviceEventHandler;
cam->RegisterEvent(allDeviceEventHandler);
```

Grabbing Images

You can grab images using the GetNextImage() function. This function returns an image pointer for the current image. The image pointer should be released whenever you are done with the image. Image pointer, being a smart pointer, is automatically released when set to null or when out of scope.

Image Acquisition (C++)

```
// Begin acquiring images
camBeginAcquisition();>-

ImagePtr pResultImage = cam->GetNextImage();

pResultImage->Release();
```

Grab Result

In almost all cases, you should check to see if the grabbed image has any errors. To do so, you need to call getImageStatus().

To check for errors in the image (C++)

```
ImageStatus imageStatus = pResultImage->GetImageStatus();
```

Available error enums

```
/** Status of images returned from GetNextImage() call. */

enum ImageStatus
{
    IMAGE_NO_ERROR = 0,
/**< Image is returned from GetNextImage() call without any errors. */
    IMAGE_CRC_CHECK_FAILED,
/**< Image failed CRC check. */
    IMAGE_INSUFFICIENT_SIZE,
/**< Image size is smaller than expected. */
    IMAGE_MISSING_PACKETS,
/**< Image has missing packets */
    IMAGE_LEADER_BUFFER_SIZE_INCONSISTENT,
/**< Image leader is incomplete. */
    IMAGE_TRAILER_BUFFER_SIZE_INCONSISTENT,
/**< Image trailer is incomplete. */
    IMAGE_PACKETID_INCONSISTENT,
/**< Image has an inconsistent packet id. */
    IMAGE_DATA_INCOMPLETE,
/**< Image data is incomplete. */
    IMAGE_UNKNOWN_ERROR
/**< Image has an unknown error. */
};
```

Image Pointer Class

The image pointer (ImagePtr) is a smart pointer that points to the image object. You can have multiple image pointers pointing to the same object. Smart pointers automatically manage the life time of the object that it points to. You can also re-use the same image pointer object.

Image pointer should always be assigned before using.

Image Pointer Usage

```
ImagePtr pResultImage;

// Retrieve the next received image
pResultImage = cam->GetNextImage();

ImagePtr duplicateImagePtr = pResultImage;
```

Image Pointer INCORRECT Usage

```
// Incorrect usage
ImagePtr illegalImage;
illegalImgae->Create( ... );
```

Image Pointer CORRECT Usage

```
// Correct usage
ImagePtr goodImage;
goodImage = Image::Create(...);
```

Error Handling

Spinnaker C++ uses a try catch block for exception handling.

Spinnaker C++ API

```
//Assuming Camera& Cam
try
{

        cam.Init();
}
catch (Spinnaker:: Exception &e)
{
//Exception handling
}
```

Loading and Saving Images

Loading and saving a raw image (.raw) from disk into Spinnaker library can be achieved via Image class's smart pointer.

Loading and Saving Images

```
int offlineImageWidth = 1280;
int offlineImageHeight = 1024;
int offlineOffsetX = 0;
int offlineOffsetY = 0;
unsigned char* offlineData;

// Allocate buffer for image
offlineData = (unsigned char*)malloc(sizeof(unsigned char) * offlineImageHeight * offlineImageWidth);

// Create empty image
ImagePtr loadImage = Image::Create(offlineImageWidth, offlineImageHeight, offlineOffsetX, offlineOffsetY,
PixelFormatEnums::PixelFormat_BayerRG8, offlineData);

FILE* inFile;

// Load image from disk into buffer (offlineData)
inFile = fopen(filename.str().c_str(), "rb");
fread(offlineData, 1, offlineImageHeight * offlineImageWidth, inFile);

// Convert image to mono8 data format
loadImage->Convert(PixelFormat_Mono8);

// Save image
loadImage->Save("offline.jpg");
```

Chunk Data

Chunk data is extra information that the camera can append to each image besides image data. Examples of chunk data include frame counter, image width, image height and exposure time.

For a listing of chunk data information supported by your camera, please refer to the camera's Technical Reference manual.

An image is comprised of:

- Leader
- Image Data
- Chunk Information (i.e., gain, exposure, image size)
- Trailer

C++ Enable Chunk Data

```
Cam->ChunkSelector
ChunkSelector.SetValue(ChunkSelectorEnums ::ChunkSelector_ExposureTime) ;

Cam->ChunkEnable.SetValue(true);

Cam->ChunkModeActive.SetValue(true);
```

C++ Retrieve Chunk Data

```
const ChunkData& chunkData = rawImage->GetChunkData();

float64_t currentExposure = chunkData.GetExposureTime();
```

Sequencer

The purpose of a sequencer is to allow you to programmatically control the acquisition parameters of an image sequence. You can define not only how the images are captured (i.e. the camera feature settings) but also when the camera transitions from one acquisition setting to another. This is akin to a state machine diagram where the states correspond to the sequencer set feature settings, and the transition among states corresponds to a particular event that triggers the state machine to move from one state to another.

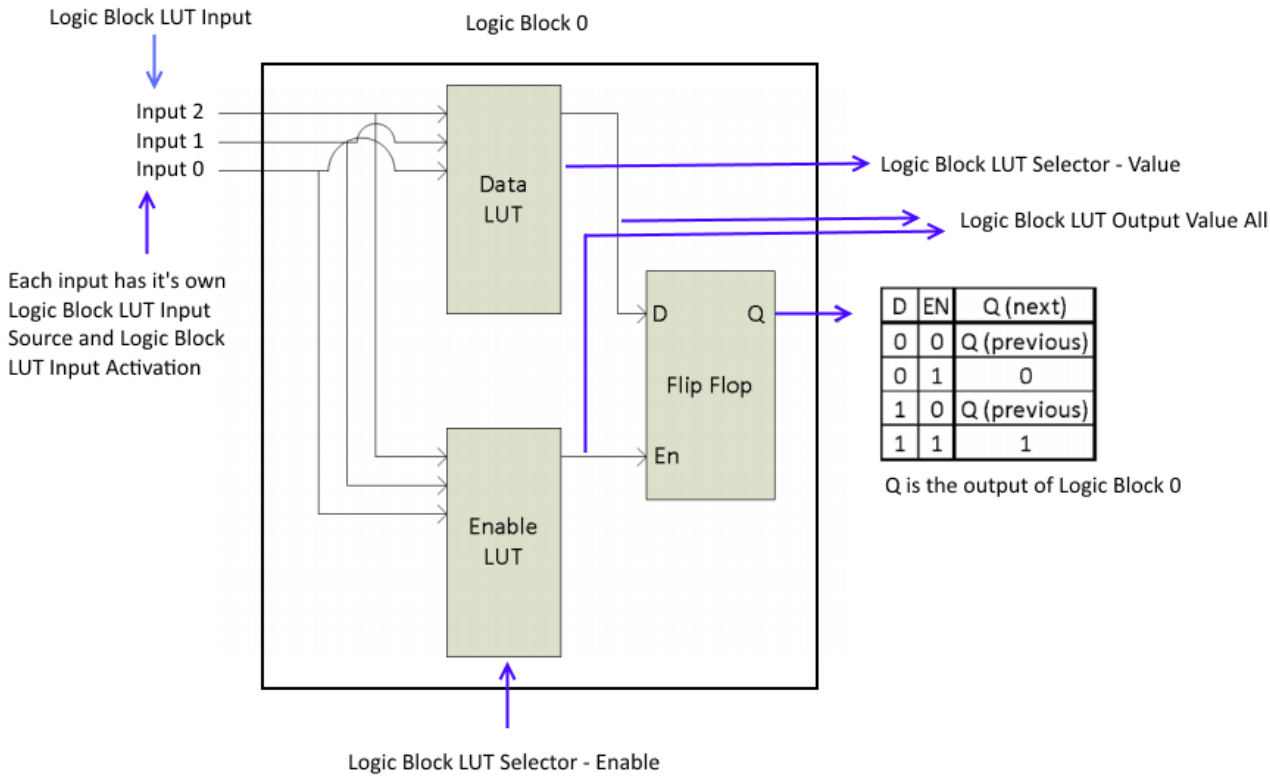
To configure sequencer on your camera, you can use SpinView's sequencer tab. Or, to programmatically configure it, you can use the C++ Sequencer source code example that is installed along with Spinnaker SDK.

Logic Block

A Logic Block is a collection of combinatorial logic and latches that allows the user to create new, custom signals inside the camera. Each Logic Block is comprised of 2 lookup tables (LUT) with programmable inputs, truth tables and a flip flop output. There is a LUT for both the D input (Value LUT) and the enable input

(Enable LUT) of the flip flop. Both LUTs have 3 inputs and thus have 8 configuration bits for their truth table.

For more information, see Using Logic Blocks.



Logging

Spinnaker supports five levels of logging:

- Error—failures that are non-recoverable (this is the default level)
- Warning—failures that are recoverable without user intervention
- Notice—information about events such as camera arrival or disconnect, camera initialize, camera start/stop, or modification of a feature
- Info—information about recurring events that are generated with every image
- Debug—information that can be used to troubleshoot the system

You can define the logging level that you want to monitor. Levels are inclusive, that is, if you monitor debug level error, you also monitor all logging levels above it.

For a complete C++ and C# example of Logging, please see Spinnaker SDK source code examples. By default, Spinnaker SDK's SpinView application saves all logging data to:

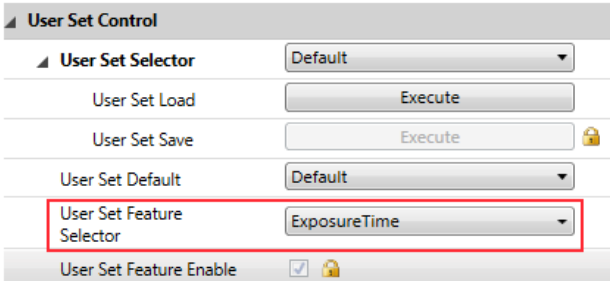
C:\ProgramData\Spinnaker\Logs

Register Logging (C++)

```
SystemPtr system = System::GetInstance();

// Register logging callback class
LogCallback callBackClass;
system>RegisterLoggingEvent((Spinnaker::LoggingEvent&)callBackClass);

// Set callback priority level
system->SetLoggingEventPriorityLevel(k_LoggingLevel);
```

```
class LogCallback : Spinnaker::LoggingEvent
{
    void OnLogEvent(LoggingEventDataPtr loggingEventDataPtr)
    {
      ...
    }

};
```

User Set

User set is an on camera non-volatile memory space that you can use to store camera properties such as exposure and gain.

To check if user set supports the feature that you want to save, you can either query the User Set Feature Selector programmatically or run SpinView: